

Exploring Innovative Machine Learning Models for Soil CO₂ Emission Forecasting

Abstract

Soil respiration represents the largest flux of carbon dioxide (CO₂) from the Earth's surface to the atmosphere. However, accurately predicting CO₂ emissions from soil respiration remains a significant challenge, and the influence of various environmental factors is not yet fully understood. This study develops machine learning models to predict CO₂ emissions from soil respiration and to evaluate the contributions of different environmental drivers. A new machine learning model, XGBoost, is introduced to estimate annual soil respiration (Rs) using spatial, temporal, climatic, and other independent variables. In addition, a deep neural network (DNN) is explored. Their performance is compared with two baseline models, Random Forest (RF) and Gradient Boosting (GB), used in previous studies. Among the models, XGBoost achieves the best performance, surpassing RF. Conversely, the DNN model struggles to produce reliable results, primarily due to the limited sample size. Additionally, SHAP (SHapley Additive exPlanations) analysis is applied to enhance model explainability, quantifying the influence of individual features on model outcomes, both at the overall level and for specific predictions.